GQE ANALYSIS PROBLEMS

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Changes to Jakob's original document are marked with blue.
Please let Alex know if you find any typos or discrepancies!

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Throughout this document there are [bracketed comments] inserted into the original problems. By and large these serve two purposes, namely to make references to previous problems that are somehow similar or related, and to make note of details in problems that are ambiguous, unclear, incorrect, or possibly just interesting. To make these comments easier to locate they are accompanied by a bullet in the right margin, as above. •

1. Continuity and Friends

Subject tags:

- [2D] = Continuity and Differentiability in $\mathbb{R}^2$
- [EIM] = Extreme Value, Intermediate Value, and Mean Value Theorems
- [LC] = Lipschitz Continuity
- [TOP] = Basic Topology
- [UC] = Uniform Continuity
- [O] = Other

Problem 1.1 (Fall 2020, New)**.** [UC] Prove that $f(x) = \ln(x)$ is not uniformly continuous on $[0, 1]$. [$f(x)$ is not defined at 0, so the interval of interested should be $(0, 1]$ instead of $[0, 1]$.] •

Problem 1.2 (Fall 2020, New)**.** [EIM] Let $f(x)$ be a continuous function on $[0, 1]$. Suppose $f(0) = f(1) = 0$. Show that there exist $a, b \in [0, 1]$ such that $f(a) = f(b)$ and $b - a = \frac{1}{2}$.

Hint. Try to introduce a suitable auxiliary function to which you may apply the intermediate value theorem.

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Problem 1.3 (Spring 2020, New)**.** [TOP] Let (X, d) be a metric space, and let $f\colon X \to X$ be a continuous function. Show that the function $g\colon X \to \mathbb{R}$ defined by $g(x) = d(x, f(x))$ is continuous using the (ε, δ)-definition of continuity.

Problem 1.4 (Fall 2019, New)**.** [UC] Prove that the function $f(x) = \tan^{-1} x$ is uniformly continuous on $[0, \infty)$. [Cf. Problem 1.1]

Problem 1.5 (Spring 2019)**.** [EIM] Let $f, g\colon [0,1] \to [0,\infty)$ be two continuous functions such that

$$\sup_{x\in[0,1]} f(x) = \sup_{x\in[0,1]} g(x).$$

Show that there exists $t \in [0,1]$ such that $f(t) = g(t)$.

Problem 1.6 (Spring 2019)**.** [O] A function $f\colon [0,1] \to \mathbb{R}$ is said to be *upper semicontinuous* if for any given $x \in [0,1]$ and $\varepsilon > 0$, there exists $\delta > 0$ such that $|y - x| < \delta$ implies $f(y) < f(x) + \varepsilon$. Show that an upper semicontinuous function f on $[0,1]$ is bounded above.

Remark. Note how if we also require $f(x) - \varepsilon < f(y) < f(x) + \varepsilon$, which is equivalent to $|f(x) - f(y)| < \varepsilon$, we have recovered the usual definition of *continuity*. In other words this is literally one half of the usual continuity.

Problem 1.7 (Fall 2018)**.** [LC] A real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ is called *Lipschitz continuous* if there exists some positive $K \in \mathbb{R}$ such that for every $x_1, x_2 \in A \subset \mathbb{R}$, $|f(x_1) - f(x_2)| \le K|x_1 - x_2|$. Let

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x^2}) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

(*a*) Find $f'(x)$.
(*b*) Is $f'(x)$ continuous?
(*c*) Is $f(x)$ Lipschitz continuous?

Problem 1.8 (Spring 2018)**.** [LC] Let a and b be real numbers with $a < b$. A function $f\colon (a,b) \to \mathbb{R}$ is called *Lipschitz* if there exists $L > 0$ such that $|f(x) - f(t)| < L|x - t|$ for all x and $t \in (a,b)$.

Let $f\colon (a,b) \to \mathbb{R}$ be differentiable. Show that f' is bounded if and only if f is Lipschitz.

Problem 1.9 (Fall 2017)**.** [TOP] Suppose that f is a real valued function on the set $S \subset \mathbb{R}$ and that c is an isolated point in S. Prove of disprove that f is continuous at c. [This is subtle.]

Problem 1.10 (Fall 2017, New)**.** [2D] Consider the following function:

$$f(x,y) = \begin{cases} (x^2 + y^2) \sin \frac{1}{x^2+y^2}, & \text{for } x^2 + y^2 \neq 0, \\ 0, & \text{for } x = 0, y = 0. \end{cases}$$

(*a*) Investigate f_x and f_y (partial derivatives of $f(x,y)$) in a neighbourhood of $(0,0)$. Are f_x and f_y continuous at $(0,0)$? Are f_x and f_y bounded in a neighbourhood of $(0,0)$?
(*b*) Is the function $f(x,y)$ differentiable at $(0,0)$?

[Cf. Problem 1.7 (a),(b)]

Problem 1.11 (Spring 2017)**.** [TOP] Suppose we define a function $d\colon \mathbb{R}^2 \to \mathbb{R}$ by

$$d(x,y) = \begin{cases} 0, & x = y \\ 1, & x \neq y. \end{cases}$$

(a) Prove that d is a metric on $\mathbb{R}$.
(b) Prove that for any $a \in \mathbb{R}$, $\{\, a \,\}$ is an open set under the metric d.
(c) Let $X = \mathbb{R}$ with the usual Euclidean distance and $Y = \mathbb{R}$ with the metric d defined above. Prove that $f\colon X \to Y$ defined by $f(x) = x$ is not continuous.

Problem 1.12 (Fall 2016, New)**.** [TOP] Let (X, d) be a metric space with the added condition that for *any* three points $x, y, z \in X$, $d(x,y) \leq \max\{d(x,z), d(y,z)\}$.

(a) Show that every triangle in X is isosceles.
(b) An open ball in X with center $u \in X$ and radius $r > 0$ is defined as $B(u;r) = \{x \in X \mid d(u,x) < r\}$. Show that every point in an open ball is a center for the open ball.

Hint. Part of your argument might include showing that if $v \in B(u;r)$, then $B(v;r) = B(u;r)$.

Problem 1.13 (Spring 2016, New)**.** [O] Find all points where the function $f(x) = |x|^p$ with $p > 0$ differentiable. How does your answer depend on p?

Problem 1.14 (Spring 2015)**.** [UC] Suppose that a function $f\colon [\frac{1}{2}, 1] \to \mathbb{R}$ satisfies

$$|f(x) - f(t)| \leq \tan(|x - t|)$$

for all x, t in the domain of f. Prove that f is uniformly continuous.

Problem 1.15 (Fall 2013, New)**.** [2D] Let

$$f(x,y) = \begin{cases} \frac{\sin(xy)}{\sqrt{x^2+y^2}}, & (x,y) \in \mathbb{R}^2 \setminus (0,0) \\ 1, & (x,y) = 0. \end{cases}$$

(a) Find the limit $\lim_{(x,y)\to 0} f(x,y)$, if it exists.
(b) Find the region in which $f(x,y)$ is continuous and the region in which $f(x,y)$ is differentiable. Indicate the reasons for your answer.

Problem 1.16 (Fall 2013)**.** [UC] Let $f(x)$ be continuous on $[0,1)$ and $\lim\limits_{x\to 1^-} f(x) = A$, where A is a real number. Prove

(a) $f(x)$ is bounded on $[0,1)$.
(b) $f(x)$ is uniformly continuous on $[0,1)$.

Problem 1.17 (Fall 2012)**.** [2D] Let

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & \text{if } (x,y) \neq (0,0), \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

Prove that $f_x(x,y)$ and $f_y(x,y)$ exist at every point of $\mathbb{R}^2$, but f is not continuous at $(0,0)$.

Problem 1.18 (Fall 2011)**.** [LC] A function $f(x)$ is called *Lipschitz continuous* on an interval $I \subset \mathbb{R}^1$ if there exists a constant L such that

$$|f(x) - f(y)| \leq L|x - y|, \qquad \text{for all } x, y \in I.$$

(a) Assume the derivative $f'(x)$ exists and is continuous on a closed finite interval $I = [a, b]$. Prove that $f(x)$ is Lipschitz continuous on $[a, b]$.

(b) Let $f(x) = x^\alpha$, $x \in [0, 1]$, where α is a nonnegative parameter. Find the range of α such that $f(x)$ is Lipschitz continuous on $[0, 1]$. Justify your answer.

[Cf. Problem 1.8.] •

Problem 1.19 (Fall 2011, New)**.** [2D] Let

$$f(x, y) = \frac{x^2 y^2}{x^4 + y^4}.$$

(a) Find the limit as $(x, y) \to (0, 0)$ along the curves $y = \sin x$ and $y = x^2$, respectively.

(b) Prove that the limit

$$\lim_{(x,y)\to(0,0)} f(x, y)$$

exists.

[Cf. Problem 1.17.] •

Problem 1.20 (Fall 2011)**.** [O] Let $F(x) = \int_a^{\beta(x)} f(x, y)\, dy$, where $\beta(x)$ is defined on a bounded interval $[a, b]$ and $f(x, y)$ is defined on a rectangle $R = [a, b] \times [a, b]$ and is continuous for all $(x, y) \in R$. Moreover, $a \le \beta(x) \le b$ for all $x \in [a, b]$.

(a) Prove $F(x)$ is continuous on $[a, b]$ if $\beta(x)$ is continuous on $[a, b]$.

(b) Assume $\beta(x)$, $\beta'(x)$, $f(x, y)$, and $\frac{\partial f(x,y)}{\partial x}$ exist and are continuous in their domains. Prove that $F(x)$ is also differentiable and

$$F'(x) = \beta'(x) f(x, \beta(x)) + \int_a^{\beta(x)} \frac{\partial f}{\partial x}(x, y)\, dy.$$

Problem 1.21 (Spring 2011, Spring 2010)**.** [UC] Let I be a bounded interval in $\mathbb{R}$. Suppose $f\colon I \to \mathbb{R}$ is uniformly continuous on I. Show that the set $f(I) \subseteq \mathbb{R}$ is bounded. [Cf. Problem 1.6.] •

(Reminder: f is *uniformly continuous* if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $x, y \in I$ satisfying $|x - y| < \delta$.)

Problem 1.22 (Fall 2010)**.** [UC] Show that the function $f(x) = x^2$ is continuous but not uniformly continuous on the interval $S = (0, \infty)$. [Cf. Problems 1.1, 1.4.] •

Problem 1.23 (Fall 2010)**.** [EIM] Suppose f and g are continuous functions at $c \in (a, b)$ and $f(c) > g(c)$. Show that there exists $\delta > 0$ such that for all $x \in (a, b)$ with $|x - c| < \delta$, we have $f(x) > g(x)$.

Problem 1.24 (Fall 2019, New)**.** [O] Prove that if a real-valued function f, defined on $[0, 1)$, is monotone and bounded, then $\lim_{x\to\infty} f(x)$ exists (as a real number, not "infinity").

2. Integrals, Taylor's Theorem (New)

Subject tags:

- [II] = Improper Integrals
- [MVT] = Mean Value Theorem for Integrals
- [TT] = Taylor's Theorem
- [O] = Other

Problem 2.1 (Fall 2021, Spring 2013, New)**.** [TT] Let $f\colon \mathbb{R} \to \mathbb{R}$ be twice continuously differentiable and suppose that $|f''(x)| \leq M$ for some $M > 0$. Find a condition on $y \in \mathbb{R}$ that ensures that the inequality

$$f\left(x - \alpha\left(f'(x) + y\right)\right) < f(x)$$

holds for all sufficiently small $\alpha > 0$, provided $f'(x) > 0$.

Hint. Use Taylor formula for $f(x+z)$, where $z = -\alpha\left(f'(x) + y\right)$.

Problem 2.2 (Fall 2021, Spring 2013, New)**.** [II] Is there an $x \in \mathbb{R}$ such that the improper integral

$$I(x) = \int_2^\infty \frac{dy}{y\sqrt{\ln(y + x^2)}}$$

converges?

Problem 2.3 (Spring 2021, New)**.** [MVT] Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous function. Show that

$$\int_0^1 f(x)x^4\,dx = \frac{1}{5}f(c), \quad \text{for some} c \in [0,1].$$

Problem 2.4 (Spring 2021, New)**.** [II] Let $f\colon [0,\infty) \to \mathbb{R}$ be a *non-negative*, uniformly continuous function. Suppose that the improper integral

$$\int_0^\infty f(x)\,dx$$

converges. Prove that $\lim_{x\to\infty} f(x) = 0$.

Hint. What would be true if $f(x)$ does not converge to zero at infinity?

Problem 2.5 (Fall 2020, New)**.** [TT] Suppose that $f(x,y)$ is a real valued continuously differentiable function of two variables such that $\|\nabla f(x,y)\| \leq 1$. Consider the surface integral

$$I = \int_S |f(x,y) - f(0,0)|\,d\sigma,$$

where S is the cone defined by $z^2 = x^2 + y^2$, $0 \leq z \leq 1$. Prove that

$$|I| \leq \frac{2\sqrt{2}}{3}\pi.$$

Hint. You may try to apply the multidimensional Taylor formula. A special case of this formula is given by

$$f(x,y) = f(a,b) + \nabla f(s,t)\cdot(x-a, y-b),$$

where $(s,t) \in \mathbb{R}^2$, $s = a + \theta(x-a), t = b + \theta(y-b)$ for some $\theta \in (0,1)$.

Problem 2.6 (Spring 2020, New). [O] Let $f(x) = \frac{1}{x}\int_0^x e^{-t^2}\,dt$, for $x > 0$. Show that $f(x)$ is a decreasing function on $(0, \infty)$.

Problem 2.7 (Spring 2020, New). [II] Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that $f(x) \geq 0$ for all x in $\mathbb{R}$. Define

$$g(x) := \sum_{n=1}^{\infty} f(x+n).$$

Suppose that $\int_{-\infty}^{\infty} g(x) < \infty$. Show that $f \equiv 0$.

Problem 2.8 (Spring 2019, New). [II]

(*a*) Prove that $\sin x \leq x$ for all $x \in \left[0, \frac{\pi}{2}\right]$.

(*b*) Determine whether the improper integral

$$\int_0^{\frac{\pi}{2}} \frac{dx}{x \sin x}$$

converges or diverges.

Problem 2.9 (Fall 2018, New). [O] Let

$$f(x) = \int_0^x \int_t^x \sin\left(1 + s^3\right)\, dsdt.$$

Write an expression for the derivative f'.

Hint. You may want to draw the domain of integration.

Problem 2.10 (Fall 2018, New). [TT] Prove that for $x > 0$,

$$\ln(1+x) < x - \frac{1}{2}x^2 + \frac{1}{3}x^3.$$

Problem 2.11 (Spring 2018, New). [MVT] Let $f(x, y)$ and $g(x, y)$ be continuous on a bounded, closed, and connected set $D \subset \mathbb{R}^2$ and $g(x, y) \geq 0$ on D. Prove the following mean-value theorem for double integral: There exists a point $(x_0, y_0) \in D$, such that

$$\int\int_D f(x,y)g(x,y)\,dA = f(x_0, y_0)\int\int_D g(x,y)\,dA.$$

Problem 2.12 (Spring 2017, New). [O] Let $k(x) = h(x) - g(x)$ where g and h are infinitely differentiable functions from $\mathbb{R}$ to $\mathbb{R}$.

(*a*) Suppose $h'(x) = g'(x)$ for all x. Characterize the possible functions $k(x)$.

(*b*) Suppose $h''(x) = g''(x)$ for all x. Characterize the possible functions $k(x)$.

(*c*) Suppose $h''(x) = g''(x)$ for all x. Suppose further that $k(0) = 0$ and $|k'(0)| < 1$. Find $\lim_{n\to\infty} k(x)^n$ for $|x| < 1$.

(*d*) Define $f^{(n)}$ to be the n-th derivative of $f\colon \mathbb{R} \to \mathbb{R}$. Suppose $h^{(n)}(x) = g^{(n)}(x)$ for all x. Suppose $k(x_1) = k(x_2) = \cdots = k(x_n) = 0$ for $x_1 < x_2 < x_3 < \cdots < x_n$. Show that $g = h$.

Problem 2.13 (Spring 2016, New). [TT] For this problem:

(*a*) Find the Taylor series for $g(x) = \frac{1}{1+x^2}$ about $x_0 = 0$.

(*b*) Find the Taylor series for $f(x) = \int_0^x \frac{1}{1+t^2}\,dt$ about $x_0 = 0$, and determine its interval of convergence,

(c) Find the value of $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)\left(\sqrt{3}\right)^{2n+1}}$.

Problem 2.14 (Fall 2015, New)**.** [II] Suppose that a function f is infinitely many times continuously differentiable on the real line, and f and all of its derivatives vanish outside of the interval $[0,1]$. Note that the above assumptions imply that $f^{(N)}(0) = f^{(N)}(1) = 0$ for each $N = 0, 1, 2, \ldots$. Here if $N > 0$, then $f^{(N)}$ denotes the derivative of order N, and if $N = 0$, then $f^{(N)}$ denotes f.

Consider the function

$$I(x) = \int_{-\infty}^{\infty} \cos(tx) f(t)\, dt.$$

Show that for each integer $N \geq 0$ there is a constant C_N such that

$$|I(x)| \leq C_N |x|^{-N}$$

for $x \neq 0$. To make estimation easier, you may assume that N is a multiple of 4.

Problem 2.15 (Fall 2015, New)**.** [TT]

(a) Evaluate the indefinite integral as an infinite series:

$$\int x \cos\left(x^3\right)\, dx.$$

(b) Approximate the definite integral

$$\int_0^1 x \cos\left(x^3\right)\, dx.$$

to within two decimal places.

Problem 2.16 (Spring 2015, New)**.** [TT] Suppose $f(x)$ is three times continuously differentiable on $[a,b]$. Define

$$I(f) = \frac{b-a}{2}\left(f(a) + f(b)\right).$$

$I(f)$ provides an approximation to the area between $f(x)$ and the x-axis over the interval $[a,b]$, i.e., $\int_a^b f(x)dx$, using a trapezoid with heights $f(a)$ and $f(b)$, and width $(b-a)$. Show that

$$\int_a^b f(x)\, dx - I(f) = -\frac{1}{12} f''(m)(b-a)^3 + O\left((b-a)^4\right),$$

when $m = \frac{a+b}{2}$.

Hint. Use Taylor formula for $f(x)$ centered at $m = \frac{a+b}{2}$ to approximate $f(a)$, $f(b)$, and $f(x)$.

Problem 2.17 (Fall 2014, Moved)**.** [II] Let $f(x)$ be a continuous real-valued function satisfying

$$\int_0^{\infty} |f(x)|\, dx < \infty.$$

(a) Show that

$$\lim_{n\to\infty} \frac{1}{n} \int_0^n x f(x)\, dx = 0.$$

(b) Find a function $f(x)$ satisfying $\int_0^{\infty} |f(x)|\, dx < \infty$ for which $\lim\limits_{x\to\infty} f(x) \neq 0$.

[Cf. Problem 2.4] •

Problem 2.18 (Spring 2014, New)**.** [O] Find the area of the region in $\mathbb{R}^2$ bounded by the curves $y = x^2$, $y = x + 6$, $x = 0$, and $x = 5$.

Problem 2.19 (Spring 2014, Moved)**.** [MVT] Let $F, G\colon [a, b] \to \mathbb{R}$ be two continuous functions on $[a, b]$ with $G(x) \geq 0$ for all $x \in [a, b]$. Prove that there exists at least one $\xi \in [a, b]$ such that

$$\int_a^b F(x)G(x)\,dx = F(\xi)\int_a^b G(x)\,dx.$$

(That is, prove the Second Mean Value Theorem for Integrals.)

Problem 2.20 (Fall 2013, New)**.** [TT] Let $f(x)$ be differentiable up to the second order on $[a, b]$. Prove

$$\left|\int_a^b f(x)\,dx - (b-a)f\left(\frac{a+b}{2}\right)\right| \leq \frac{M(b-a)^3}{24},$$

where

$$M = \max_{[a,b]}|f''(x)|.$$

Hint. Use Taylor expansion.

Problem 2.21 (Fall 2011, New)**.** [TT] Let

$$I = \int_0^1 \frac{1-\cos(2x)}{x}\,dx.$$

(a) Find a power series for the integrand.
(b) Write down the infinite series for the exact value of integral I and justify why it is convergent.
(c) Use the series in (b) to estimate I with an error less than 10^{-3}.

Problem 2.22 (Spring 2011, New)**.** [TT] Suppose a, b, c, and d are positive real numbers with $a > d$. Use an appropriate Taylor polynomial to show that if

$$\left(c - \frac{cd}{a}\right)^2 \gg \frac{4bc}{a},$$

then the expression

$$z = \sqrt{\left(c - \frac{cd}{a}\right)^2 + \frac{4bc}{a}}$$

can be approximated by

$$z \approx c - \frac{cd}{a} + \frac{2b}{a-d}.$$

Problem 2.23 (Fall 2009, New)**.** [TT] If $x > 0$, use Taylor's Theorem to prove that

$$2 + \frac{x}{4} - \frac{x^2}{64} \leq \sqrt{4+x} \leq 2 + \frac{x}{4}.$$

3. Multivariate Calculus

Subject tags:

- [DGS] = Divergence Theorem, Green's Theorem, and Stoke's Theorem
- [DI] = Double Integrals Computation
- [IIT] = Inverse Function Theorem and Implicit Function Theorem
- [OPT] = Optimization
- [O] = Other

Problem 3.1 (Fall 2021, New)**.** [DGS] Evaluate the line integral $I = \int \boldsymbol{F} \cdot d\boldsymbol{r}$ of the vector field

$$\boldsymbol{F}(x, y, z) = \langle x^2 z + e^{\sqrt{x+2}}, x, y^3 \rangle,$$

where the curve C is the circle of the unit radius centered at the origin in the xy-plane. Assume that C is positively oriented.

Problem 3.2 (Spring 2021, New)**.** [DGS] In a three dimensional space with Euclidean coordinates x, y, z consider the trapezoid in the yz-plane with vertices $(0, 0, 0)$, $(0, 1, 0)$, $(0, 1, 1)$, and $(0, 0, 2)$. When this trapezoid is revolved around the z-axis, it sweeps a region $R \subset \mathbb{R}^3$. Let S denote the boundary of R. Evaluate the surface integral

$$\int_S \boldsymbol{f} \cdot \boldsymbol{n} \, d\sigma$$

where $\boldsymbol{f} = (-z^2 + x, -y, z)$, and $\boldsymbol{n}$ is the outward pointing unit normal to S.

Problem 3.3 (Spring 2019)**.** [DGS] Use Green's theorem and a suitable change of variables to evaluate

$$\oint_C (2x^3 - y^3) \, dx + (x^3 + y^3) \, dy$$

where C is the counter-clockwise oriented circle $x^2 + y^2 = 1$.

Problem 3.4 (Fall 2017)**.** [DGS] Consider the vector field in $\mathbb{R}^3$ defined by

$$F(x, y, z) = M \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

where M is a 3×3 symmetric matrix with real entries. Compute the following integrals and express the results in terms of M if possible.

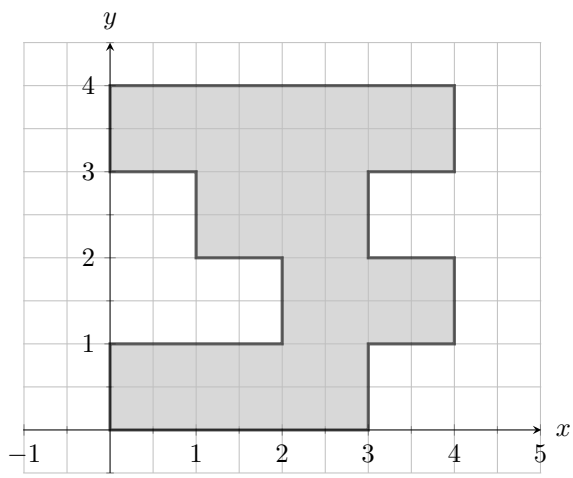
(a) $\oiint_S F \cdot dS$, where S is the surface of the unit sphere in $\mathbb{R}^3$ centred at the origin.

(b) $\oint_C F \cdot dr$, where C is the shortest path curve along the unit sphere from $(0, 0, 1)$ to $(0, 1, 0)$ to $(1, 0, 0)$ to $(0, 0, 1)$.

Problem 3.5 (Spring 2017)**.** [DGS]

(a) State the divergence theorem relating an integral over a region $\Omega \subset \mathbb{R}^2$ to an integral over the boundary of that region $\partial\Omega$.

(b) Let $\vec{V} = (v_1(x, y), v_2(x, y)) = (x, y)$. Compute $\int_{\partial\Omega} \vec{n} \cdot \vec{V} \, d\ell$ where Ω is the shaded region in the figure below.



Problem 3.6 (Fall 2016, New)**.** [OPT] Find all points on the surface $xy - z^2 = 1$ which are closest to the origin. Here we take x, y, x to be the cartesian coordinates in $\mathbb{R}^3$.

Problem 3.7 (Fall 2016, New)**.** [IIT] Consider the surface $z(w, x, y) = 0$. Determine $\frac{\partial y}{\partial x}$ as a function of first partial derivatives of z when the quantity $x^2 w$ is held constant. Assume that all first partial derivatives are non zero on the open set of interest.

Hint. Use the same ideas as used to prove the implicit function theorem.

Problem 3.8 (Spring 2016)**.** [DGS] Evaluate the line integral in Stokes' Theorem to determine the value of the surface integral

$$\iint_S (\nabla \times \boldsymbol{F}) \cdot \boldsymbol{n}\, dS,$$

where $\boldsymbol{n}$ is the outward unit normal, $\boldsymbol{F} = \langle \boldsymbol{x}^2 - \boldsymbol{z}^2, \boldsymbol{y}^2, \boldsymbol{xz} \rangle$, and S is the hemisphere $x^2 + y^2 + z^2 = 4$ for $y \geq 0$.

Problem 3.9 (Fall 2015)**.** [DGS] Consider the plane with the unit normal vector $\boldsymbol{n} = \frac{1}{\sqrt{3}}\langle 1, 1, 1 \rangle$, and a simple closed smooth curve C whose orientation is positive with respect to $\boldsymbol{n}$. Show that the area of the plane region enclosed by C is given by the line integral

$$\int_C \boldsymbol{G} \cdot d\boldsymbol{r}$$

where the vector field $\boldsymbol{G} = \sqrt{3}\langle z, x, y \rangle$. Notation: $\boldsymbol{a} = \langle a, b, c \rangle$ indicates that a, b, c are Cartesian components of the vector $\boldsymbol{a} \in \mathbb{R}^3$. [There is at least one typo in this problem: as stated it is incorrect.]

Problem 3.10 (Fall 2015, New)**.** [IIT] Can the equation $\left(x^2 + y^2 + 2z^2\right)^{\frac{1}{2}} = \cos z$ be solved uniquely for y in terms of x and z near $(0, 1, 0)$?

Problem 3.11 (Spring 2015, New)**.** [DI] Evaluate the double integral

$$\int\int_R e^{x^2 - 2xy + 10y^2}\, dA,$$

where $R = \{(x, y) \in \mathbb{R}^2 \mid x^2 - 2xy + 10y^2 \leq 1\}$.

Problem 3.12 (Spring 2015, New)**.** [IIT] Consider a mapping $f : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$f(u, v) = \left(u^2 + uv - v^2, u - v\right).$$

Prove that f is locally invertible near the point $(1, 0) = f(1, 1)$.

Problem 3.13 (Fall 2014, New)**.** [IIT] Define $f(x, y) = x^4 + y^2 - 6y + 9$. We will look at implicit functions defined by $f(x, y) = x^4 + y^2 - 6y + 9 = 1$.

(a) State the implicit function theorem in the case of $f : \mathbb{R}^2 \to R$.
(b) At what points on the x-axis can we (locally) find a function g such that $f(x, g(x)) = 1$?

Problem 3.14 (Fall 2014)**.** [DGS] Find the simple closed curve C that maximises the line integral

$$\int_C (x^2 y)\, dx + (x - xy^2)\, dy$$

and find the maximum value of the line integral.

Problem 3.15 (Spring 2014)**.** [DGS] Let $\vec{V} = (V_1, V_2) = (-y, x)$ be a vector field in $\mathbb{R}^2$. Let $\Omega = [0, L] \times [0, L]$, the square region with side-length L. Let $f \colon \mathbb{R}^2 \to \mathbb{R}$ be any smooth (infinitely differentiable) function on $\mathbb{R}^2$. Denote the boundary of Ω by $\partial\Omega$.

(a) State Stokes theorem relating path integrals of a vector field around a closed curve to integrals of the curl of the same vector field.
(b) Compute $\int_{\partial\Omega} \vec{V} \cdot dr$ directly and also by using Stokes theorem.
(c) Suppose that you are given a different vector field $\vec{V} = \nabla f$. What can you say about $\int_{\partial\Omega} \vec{V} \cdot dr$?

Problem 3.16 (Spring 2013)**.** [DGS] Find all a, b such that the vector field

$$F(x, y) = \left(\frac{x + ay}{x^2 + y^2}, \frac{bx + y}{x^2 + y^2}\right)$$

is conservative in $\Omega = \{\, (x, y) \mid 1 < x^2 + y^2 < 4 \,\}$. Recall that a vector field is conservative if

$$\int_C F \cdot dr = 0$$

for each closed curve $C \subset \Omega$.

Problem 3.17 (Spring 2013, New)**.** [DGS] Let $\phi(x, y, z) = x + 2y + z$. Evaluate

$$I = \int_S \phi \frac{\partial \phi}{\partial n}\, dS,$$

where S is the sphere $x^2 + y^2 + z^2 = 1$ and $\frac{\partial}{\partial n}$ denotes the normal derivative.

Problem 3.18 (Spring 2012, New)**.** [DI] Let D be the bounded region in the first quadrant of $\mathbb{R}^2$ enclosed by the parallel lines $y = x$ and $y = x + 1$ together with the curves $xy = 1$ and $xy = 2$. Evaluate

$$\int\int_D (y^2 - x^2)\, dxdy.$$

Problem 3.19 (Fall 2011, New)**.** [DI] Calculate the double integral

$$\int\int_R xy\,dxdy,$$

where R is the region bounded by $x^2+y^2=4$, $x^2+y^2=9$ and $x^2-y^2=1$, $x^2-y^2=4$, $x\geq 0$, $y\geq 0$.

Hint. Use the transformation $u=x^2+y^2$, $v=x^2-y^2$.

Problem 3.20 (Spring 2011, New)**.** [DI] Evaluate the integral

$$\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}} y(x^2+y^2)^{\frac{7}{2}}\,dydx$$

Problem 3.21 (Spring 2011, New)**.** [OPT] Find the maximum value of the function

$$f(x,y,z)=\ln(x)+\ln(y)+3\ln(z)$$

on that portion of the sphere

$$g(x,y,z)=x^2+y^2+z^2=10$$

in the octant $x>0$, $y>0$, $z>0$.

Problem 3.22 (Spring 2011)**.** [DGS] In a three dimensional space with Euclidean coordinates x, y, z consider the rectangle with vertices $(1,0,0)$, $(2,0,0)$, $(2,3,0)$, and $(1,3,0)$. When this rectangle is revolved around the y-axis, it sweeps a "rectangular torus" T. Evaluate the surface integral

$$\int_T \boldsymbol{f}\cdot\boldsymbol{n}\,dS,$$

where $\boldsymbol{f}(x,y,z)=(-y,x,z)$, $\boldsymbol{n}$ is the outward unit normal to T, and dS is the surface area element on T. [Cf. Problem 3.2]

Problem 3.23 (Fall 2010, New)**.** [OPT] Consider the function $f(x,y,z)=z+e^x\cos(\pi y)$, where $x,y,z\in\mathbb{R}$.

(a) Compute ∇f and find any stationary points of f.
(b) Compute the general directional derivative of f.
(c) Let l be the line described parametrically as the set of points

$$(x,y,z)=(1,2,3)+t(\pi,1,0),\quad t\in\mathbb{R}.$$

Find any stationary points (x^*,y^*,z^*) of the one-dimensional function given by f constrained to the line l.

Problem 3.24 (Spring 2010)**.** [DGS] Prove that

$$\oint_C P\,dx+Q\,dy=2\pi$$

where

$$P=-\frac{y}{x^2+y^2}\quad\text{and}\quad Q=\frac{x}{x^2+y^2}$$

and C is a simply closed differentiable curve oriented counterclockwise with $(0,0)$ as an interior point of the region bounded by C.

Hint. Green's theorem may be used.

Problem 3.25 (Fall 2009, New)**.** [O] Let $f:\mathbb{R}^3\to\mathbb{R}$ be defined by $f(x,y,z)=x^2y+xz^2-3$.

(a) Find a vector in $\mathbb{R}^4$ perpendicular to the surface $w = f(x, y, z)$ at the point $(x, y, z, w) = (1, 2, 1, 0)$.
(b) Find the equation of the hyperplane tangent to the surface $w = f(x, y, z)$ at the point $(x, y, z, w) = (1, 2, 1, 0)$.
(c) Find the directional derivative for f at the point $(1, 2, 1)$ in the direction $\langle 1, 0, -1\rangle$.

Problem 3.26 (Fall 2009, New)**.** [O] Assume the vector field $\boldsymbol{F}$ is defined by $\boldsymbol{F} = \langle e^y, xe^y, (z+1)e^z\rangle$. If $C(t)$ is the parameterized curve $C(t) = \langle t^2, 2t^3 + 1, t\rangle$ for $0 \le t \le 1$, evaluate the line integral $\int_C \boldsymbol{F} \cdot d\boldsymbol{r}$.

Hint. First show that $\boldsymbol{F}$ is conservative and find a potential function for $\boldsymbol{F}$.

Problem 3.27 (Spring 2009)**.** [DGS] Let

$$I = \int_\Gamma \left(e^{2y}\cos 3x + \alpha y^2\right)\, dx + \left(-3 + 2xy + \beta e^{2y}\sin 3x\right)\, dy,$$

where Γ is a piecewise smooth curve from $(0, 1)$ to $(\pi/2, 0)$ and α and β are constants.

(a) Find α and β such that I is independent of the choice of Γ.
(b) Evaluate I for the choice of α and β in (a).

Problem 3.28 (Spring 2009, New)**.** [OPT] Find the minimum distance from the surface to the cone $z = \sqrt{x^2 + y^2}$ to the point $(4, 0, 0)$.

4. Sequences and Series

Subject tags:

- [PSR] = Power Series (should be merged with Taylor's Theorem problems)
- [NSQ] = Numerical Sequences
- [NSR] = Numerical Series
- [FSQ] = Sequences of Functions
- [FSR] = Series of Functions
- [O] = Other

Problem 4.1 (Fall 2021, New)**.** [NSR] Does the series $\sum_{k=2}^{\infty} \log\left(\frac{k(k+2)}{(k+1)^2}\right)$ converge?

Hint. Try to simplify the n-th partial sum algebraically.

Problem 4.2 (Fall 2021, New)**.** [NSQ] State the definition of a Cauchy sequence of real numbers. State the definition of a uniformly continuous function on $\mathbb{R}$. Next, consider a sequence $\{x_n\}_{n=1}^{\infty} \subset X \subset \mathbb{R}$, and suppose that the function $f\colon \mathbb{R} \to \mathbb{R}$ is uniformly continuous on X. Assuming that the sequence $\{x_n\}_{n=1}^{\infty}$ is Cauchy, prove that the sequence $\{f(x_n)\}_{n=1}^{\infty}$ is Cauchy.

Problem 4.3 (Spring 2021, New)**.** [FSQ] Prove that $\lim_{n\to\infty} \cos\left(x + \frac{1}{n}\right) = \cos(x)$ uniformly on $\mathbb{R}$.

Problem 4.4 (Spring 2021, New)**.** [NSR] Suppose $a_k \geq 0$ for all $k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} \frac{a_k}{k} < \infty$. Show that

$$\lim_{n\to\infty} \sum_{k=1}^{\infty} \frac{a_k}{n+k} = 0.$$

Problem 4.5 (Fall 2020, New)**.** [NSQ] A sequence $\{a_n\}_{n=1}^{\infty}$ of real numbers is called *contractive* if there exists $\alpha \in (0,1)$ such that

$$|a_{n+2} - a_{n+1}| \leq \alpha |a_{n+1} - a_n|$$

for all integers $n \geq 1$. Prove that a contractive sequence is Cauchy.

Problem 4.6 (Fall 2020, New)**.** [FSQ] Let $A \subset \mathbb{R}$ and $\{f_n(x)\}$ be a sequence of functions on A. Suppose for each n, there exists a constant $M_n > 0$ such that $|f_n(x)| \leq M_n$ for all $x \in A$. Prove that if f_n converges to f uniformly on A, then there exists a constant $M > 0$ such that $|f_n(x)| \leq M$ for all n and all $x \in A$.

Problem 4.7 (Spring 2020, New)**.** [NSQ] Let $a_n = \int_1^n \frac{\sin(x)}{\sqrt{x}}\, dx$ for each number n. Prove that $\lim_{n\to\infty} a_n$ exists.

Hint. Use integration by parts.

Problem 4.8 (Fall 2019, New)**.** [NSQ] Use the (N/ε) definition of a limit of a sequence of real numbers to prove the following statement: If the sequence of reals $\{a_n\}$ converges to A and there exists n_1 such that $a_n > 0$ for all $n > n_1$, then $A \geq 0$.
Note: By (N/ε) definition, we mean "given $\varepsilon > 0$, there exists N such that for all $n > N \ldots$". No credit will be given if this definition is not used.

Problem 4.9 (Fall 2019, New)**.** [NSR] Investigate the convergence of the series

$$\sum_{n=1}^{\infty} \frac{5n}{3n^3+n^2}\sqrt{\frac{n}{\log(n+1)}}.$$

Hint. $\log(n+1) \geq \log(2)$.

Problem 4.10 (Fall 2019, New)**.** [NSQ] Let f be a continuous function on $[a,b]$ and let $M = \max_{x\in[a,b]}|f(x)|$. Show that

$$\lim_{n\to\infty}\left(\int_a^b |f(x)|^n\,dx\right)^{\frac{1}{n}} = M.$$

Problem 4.11 (Spring 2019)**.** [PSR] Let α be any irrational real number and define c_n to be the nth value after the decimal point of α; for $n \geq 1$, c_n will be in the set $\{\,0,1,2,3,4,5,6,7,8,9\,\}$. Determine the radius of convergence of

$$\sum_{n=1}^{\infty} c_n x^n.$$

You are not required to test whether the series converges at both endpoints.

Problem 4.12 (Fall 2018)**.** [FSQ] Let $f_n(x) = nx(1-x^2)^n$, for all $x \in [0,1]$, and $n = 1,2,\ldots$.

(a) Find $\lim\limits_{n\to\infty} f_n(x)$.

(b) Does the sequence $\{\,f_n\,\}$ converge uniformly on $[0,1]$?

Problem 4.13 (Spring 2018)**.** [NSQ] Let $\{\,a_n\,\}_{n=1}^{\infty}$ be a real-valued sequence where

$$\lim_{n\to\infty} a_n = 0.$$

For $n = 1,2,3,4,\ldots$, define

$$b_n = \frac{a_1+a_2+\cdots+a_n}{n}.$$

Prove that $\lim\limits_{n\to\infty} b_n = 0$.

Problem 4.14 (Spring 2018)**.** [PSR] Let $f(x)$ be defined by

$$f(x) = \int_0^{x^2} \sqrt{1+t^3}\,dt.$$

(a) Find $f'(x)$.

(b) Express $f(x)$ as a power series centered at $x=0$ and indicate the radius of convergence of the power series. You may use the binomial series

$$(1+t)^k = \sum_{n=0}^{\infty}\binom{k}{n}t^n; \quad \binom{k}{n} = \frac{k(k-1)(k-2)\cdots(k-n+1)}{n!}$$

and the fact that the radius of convergence of this series is 1.

Problem 4.15 (Fall 2017)**.** [NSQ] Let $x_1 = 1$ and define $x_{k+1} = \sqrt{2x_k}$ for $k \in \mathbb{N}$. Prove that $\{\,x_k\,\}_{k=1}^{\infty}$ converges and find its limit.

Problem 4.16 (Spring 2017)**.** [NSR] Show that the series $\sum_{n=1}^{\infty} n^2 3^{-n}$ converges to a value less than 2 using steps (a), (b), and (c) below.

(a) Define $a_n = n^2 3^{-n}$. Show that for $n \geq 3$, $\frac{a_{n+1}}{a_n} \leq \frac{2}{3}$. You may use this result to conclude $0 < a_n \leq a_3 \left(\frac{2}{3}\right)^{n-3}$.

(b) For $|a| < 1$, determine $S = \sum_{k=0}^{\infty} a^k = 1 + a + a^2 + a^3 + \ldots$ and justify your answer.

(c) Show that $\sum_{n=1}^{\infty} n^2 3^{-n} = \frac{1}{3} + \frac{4}{9} + \sum_{n=3}^{\infty} n^2 3^{-n} < 2$.

Problem 4.17 (Fall 2016)**.** [NSQ] Let $a_1 = \sqrt{2}$ and $a_{k+1} = \sqrt{2 + \sqrt{a_k}}$, $k = 1, 2, \ldots$. Show that the sequence $\{ a_k \}$ is convergent. [Cf. Problem 4.15.]

Problem 4.18 (Fall 2016)**.** [FSR] Find the largest interval (in x) where the series $\displaystyle\sum_{n=1}^{\infty} x e^{-nx}$ converges absolutely.

Problem 4.19 (Spring 2016)**.** [NSQ] Let α be any number in $(0, 1)$. Consider the sequence $\{ x_n \}_{n=0}^{\infty}$ that satisfies

$$x_{n+1} = \alpha x_n + (1 - \alpha) x_{n-1}.$$

(a) Prove that $\{ x_n \}_{n=0}^{\infty}$ is Cauchy.

(b) Derive an expression for $x_n - x_{n-1}$ in terms of α, x_0, and x_1.

(c) Derive an expression for $\lim\limits_{n\to\infty} x_n$ in terms of α, x_0, and x_1.

[Cf. Problem 2.4]

Problem 4.20 (Fall 2015)**.** [NSQ] Let $\{ x_n \}$ be a Cauchy sequence in a metric space X, and let $\{ x_{n_k} \}$ be a subsequence. Prove that $x_{n_k} \to x$ as $k \to \infty$ implies that $x_n \to x$ as $n \to \infty$.

Problem 4.21 (Spring 2015)**.** [FSQ] Consider a sequence of functions

$$f_n(x) = \frac{nx}{\cos\left(\frac{x}{n}\right) + n^2 x}$$

defined on $D = [0, 1]$.

(a) Find the function f such that $f_n \to f$ pointwise on D;

(b) Prove that $f_n \to f$ uniformly.

Problem 4.22 (Fall 2014)**.** [FSQ]

(a) Is the function $f(x) = \sin x$ uniformly continuous on $\mathbb{R}$? Justify your answer.

(b) Show that the sequence $\{ \sin n \}_{n=1}^{\infty}$ is divergent.

(c) Is there a convergent subsequence in (b)? Explain why or why not.

Problem 4.23 (Spring 2014)**.** [FSR] Let $a_n = nxe^{-nx^2}$, for $n = 1, 2, \ldots$, and $0 \leq x \leq 1$.

(a) Determine the limit $L(x) = \lim\limits_{n\to\infty} a_n(x)$.

(b) Determine if the convergence to $L(x)$ in part (a) is uniform.

Hint. First show that $\max\limits_{x\in[0,1]} a_n(x) = \sqrt{n/(2e)}$ at $x = 1/\sqrt{2n}$.

(c) Determine $I_1 = \lim\limits_{n\to\infty} \int_0^1 a_n(x)\, dx$, and $I_2 = \int_0^1 \lim\limits_{n\to\infty} a_n(x)\, dx$, and explain why your answer to part (b) can be used to justify your values for I_1 and I_2.

Problem 4.24 (Fall 2013, Spring 2010). (*a*) [NSQ] Let $a_1, a_2, \ldots, a_k$ be k non-negative real numbers. Find the limit

$$\lim_{n\to\infty} (a_1^n + a_2^n + \ldots + a_k^n)^{1/n}, \quad \text{if it exists.}$$

[Cf. Problem 4.10]

(*b*) [PSR] Find the radius of convergence for the following power series:

$$\sum_{n=1}^{\infty} \frac{n!}{n^n} x^n.$$

Problem 4.25 (Spring 2013). [NSQ] Let a_1 and a_2 be real numbers with $0 < a_1 < a_2$. For $j \geq 3$ define

$$a_j = a_{j-1} + a_{j-2}.$$

Show that the sequence $\{ a_j \}$ cannot converge to a finite limit.

Problem 4.26 (Spring 2013, Fall 2009). [NSR] Suppose that b_j, $j = 1, 2, 3, \ldots$, are positive real numbers such that the series $\sum_{n=1}^{\infty} b_n$ converges. Prove that $\sum_{n=1}^{\infty} \frac{2 - b_n}{3 - b_n^2}$ diverges but $\sum_{n=1}^{\infty} \arctan b_n$ converges.

Problem 4.27 (Fall 2012). [FSQ] Let $f_n(x) = 3x^{2n}$, $x \in [0, 1]$. Show that:

(*a*) Sequence $\{ f_n \}_{n=1}^{\infty}$ converges pointwise for every $x \in [0, 1]$.

(*b*) Sequence $\{ f_n \}_{n=1}^{\infty}$ is not uniformly converging in $[0, 1]$.

Recall: $\{ f_n \}_{n=1}^{\infty}$ is uniformly converging to f_o in $[0, 1]$ if for any $\varepsilon > 0$ there exists an N such that: $n > N$ implies $|f_n(x) - f_o(x)| < \varepsilon$ for all $x \in [0, 1]$.

Problem 4.28 (Fall 2012). [NSR] If $\sum a_n$ converges and if $\{ b_n \}$, $b_n > 0$, $n = 1, 2, \ldots$ is monotonically converging to 0, prove that $\sum a_n b_n$ converges.

Problem 4.29 (Fall 2012). [NSQ] Let f be defined on $\mathbb{R}$. Assume that for some $x_0, L \in \mathbb{R}$

$$\lim_{x\to x_0} f(x) = L. \tag{1}$$

Prove or disprove that (1) is equivalent to the following statement: *For every sequence* $\{ x_k \}$ *such that* $x_k \neq x_0$, $k = 1, 2, \ldots$ *and* $\lim_{k\to\infty} x_k = x_0$ *we have:*

$$\lim_{k\to\infty} f(x_k) = L.$$

Problem 4.30 (Spring 2012, New). [O] Show that for all integers $p \geq 0$, $\lim_{x\to 0^+} x^{-p} e^{-\frac{1}{x}} = 0$.

Problem 4.31 (Spring 2012). [FSQ] Let L^∞ denote the collection of all bounded real-valued functions defined on $[0, \infty)$. For $f, g \in L^\infty$ define

$$d(f, g) = \sup_{x\geq 0} |f(x) - g(x)|.$$

(*a*) Show that d satisfies the requirements of being a metric.

(*b*) Assume that $f \in L^\infty$ and for all positive integers n, $f_n \in L^\infty$. Show that $\lim_{n\to\infty} d(f_n, f) = 0$ if and only if f_n converges to f uniformly on $[0, \infty)$ as $n \to \infty$.

(c) For positive integers n, define

$$f_n(x) = \begin{cases} 1, & x \in [n, n + \frac{1}{n}], \\ 0 & \text{otherwise.} \end{cases}$$

It is true that for all $x \in [0, \infty)$, $f_n(x) \to 0$ as $x \to \infty$. Determine, with justification, whether the sequence $\{ f_n \}$ converges to the constant zero function uniformly.

Problem 4.32 (Fall 2010)**.** [NSR] Show that the following series converges and find its limit:

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \ldots + \frac{1}{(3n-2)(3n+1)} + \ldots.$$

Problem 4.33 (Spring 2010)**.** [NSR] Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent series. Prove that if $k \geq 1$, then $\sum_{n=1}^{\infty} |a_n b_n|^k$ converges.

Problem 4.34 (Spring 2010)**.** [NSR] Show that if $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} c_n = 1$ then

$$\lim_{M\to\infty} \frac{1}{M} \sum_{n=1}^{M} c_n a_n = a.$$

Problem 4.35 (Spring 2009)**.** [NSR] If the series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ of real numbers converge absolutely, show that

(a) $\sum_{n=0}^{\infty} a_n^2$ converges.
(b) $\sum_{n=0}^{\infty} a_n b_n$ converges absolutely.

[Cf. Problem 4.28, Problem 4.33.]

5. Convexity and Other Inequalities

Problem 5.1 (Spring 2020, New)**.** Let $a_i, b_i \in \mathbb{R}$ for $i = 1, 2, \ldots, n$. Show that

$$\left(\sum_{i=1}^{n}(a_i + b_i)^2\right)^{\frac{1}{2}} \leq \left(\sum_{i=1}^{n} a_i^2\right)^{\frac{1}{2}} + \left(\sum_{i=1}^{n} b_i^2\right)^{\frac{1}{2}}$$

Problem 5.2 (Fall 2018)**.** Let $g\colon \mathbb{R} \to \mathbb{R}$ be Riemann integrable. Also let g be convex and differentiable. Recall that g is convex if $g(\lambda x + (1-\lambda)y) \leq \lambda g(x) + (1-\lambda)g(y)$ for each $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$.

(a) Prove that $g(x) \geq g(y) + g'(y)(x - y)$ for all $x, y \in \mathbb{R}$.

(b) Using (a), prove that

$$\int_0^1 g(f(x))\,dx \geq g\left(\int_0^1 f(x)\,dx\right).$$

Problem 5.3 (Spring 2018)**.** Suppose that a function $f\colon [a, b] \to \mathbb{R}$ is twice continuously differentiable and convex. Recall that f is *convex* if

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

for each $x, y \in [a, b]$ and each $\lambda \in [0, 1]$.

Let $x_0 \in (a, b)$ be arbitrarily chosen and let $g(x)$ be the linear function whose graph is tangent to the graph of $y = f(x)$ at the point $(x_0, f(x_0))$.

(a) Use the fact that

$$f''(x) = \lim_{h \to 0} \frac{1}{h^2}\left(f(x+h) + f(x-h) - 2f(x)\right)$$

to prove $f''(x) \geq 0$ for all $x \in (a, b)$. [The converse is also true. See Problem 5.5.] •

(b) Prove that the graph of f lies above the tangent line $y = g(x)$, that is,

$$f(x) \geq g(x)$$

for all $x \in [a, b]$. [Cf. Problem 5.2 (a).] •

Problem 5.4 (Spring 2017)**.** Let $f\colon \mathbb{R} \to \mathbb{R}$ be a function and let X be a subset of $\mathbb{R}^2$.

(a) Define what it means for f to be convex.

(b) Define what it means for X to be convex.

(c) Define $\operatorname{epi}(f) = \{\,(x, y) \mid y \geq f(x)\,\}$. Describe the relationship between $\operatorname{epi}(f)$ and the graph of $y = f(x)$ in $\mathbb{R}^2$.

(d) Prove that if $\operatorname{epi}(f)$ is a convex set, then f is a convex function. [The converse is also true.] •

Problem 5.5 (Spring 2016)**.** Let $f(x)$ be a real-valued, continuously differentiable function on $[0, +\infty)$ such that $f(x)$ is second-order differentiable on $(0, +\infty)$. Prove that if $f''(x) > 0$ for all $x > 0$, then

$$F(x) = \frac{1}{x}\int_0^x f(t)\,dt$$

is strictly convex on $(0, +\infty)$.

Hint. Show $F''(x) > 0$ for all $x > 0$.

Problem 5.6 (Spring 2012)**.** A function $f\colon \mathbb{R} \to \mathbb{R}$ is said to be *convex* on an interval $I = [s, t]$ if for every $\lambda \in [0, 1]$, $f(\lambda s' + (1-\lambda)t') \le \lambda f(s') + (1-\lambda)f(t')$ for any s', t' satisfying $s \le s' \le t' \le t$. Show that if f is convex on $I = [a, c]$ then for any b where $a < b < c$,

$$f(a - b + c) \le f(a) - f(b) + f(c).$$

Problem 5.7 (Fall 2010)**.** Let $f\colon \mathbb{R} \to \mathbb{R}$ be a monotone function that satisfies

$$f(x + y) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$. Show that $f(x) = ax$ where $a = f(1)$.

Related Problem. A function is said to be *concave* if $f(\lambda x + (1-\lambda)y) \ge \lambda f(x) + (1-\lambda)f(y)$ for all $\lambda \in [0, 1]$ and x and y in the domain of f. [Cf. convex.]

Show that if a function $f\colon \mathbb{R} \to \mathbb{R}$ is both convex and concave, then it is affine (i.e., $f(x) = ax + b$ for some a and b). [Affine is the fancy (and correct) word for what we often call 'linear' in introductory maths classes.]

Related Problem. Let $f\colon \mathbb{R} \to \mathbb{R}$ be differentiable such that for any $x, y \in \mathbb{R}$, $f(x + y) = f(x)f(y)$. Show that f is monotone.

Hint. A (maybe) easier problem: suppose there exists some a such that $f(a) \ne 0$ and some b such that $f'(b) > 0$. Show that f is strictly increasing. Now generalise.

Problem 5.8 (Fall 2009, New)**.** [This problem should probably be in a different section. It is not directly related to convexity or requires to invoke some inequality. However, you can use some properties of convex functions here.] Assume $p, q \in \mathbb{R}$ with $p > 1$, $q > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$. Assume also that $u, v \in \mathbb{R}$ with $u \ge 0$, $v \ge 0$, and $u + v \ne 0$. Consider the function $f : [0, 1] \to \mathbb{R}$ defined by

$$f(t) = ut + v\left(1 - t^q\right)^{\frac{1}{q}}, \quad t \in [0, 1].$$

Prove that $f(t)$ has a unique maximum at the point

$$t^* = \left(\frac{u^p}{u^p + v^p}\right)^{\frac{1}{q}}.$$

Problem 5.9 (Spring 2009)**.** Recall that a function $f\colon \mathbb{R} \to \mathbb{R}$ is called convex if for any $x_1 \le x_2$ and $0 \le \alpha \le 1$, it follows that $f(x_\alpha) \le y_\alpha$ where $x_\alpha = \alpha x_1 + (1-\alpha)x_2$ and $y_\alpha = \alpha f(x_1) + (1-\alpha)f(x_2)$. Suppose, in addition, that f is differentiable on $\mathbb{R}$. Show that f is convex if and only if its derivative f' is nondecreasing.

Hint. Observe that $f(x_\alpha) = \alpha f(x_\alpha) + (1-\alpha)f(x_\alpha)$ and apply the mean value theorem twice.

6. Clever Computations

Problem 6.1 (Fall 2017)**.** Compute $\sqrt{1.2}$ without using a calculator to at least three correct decimal places. All steps of your calculation should be shown and the method explained.

Problem 6.2 (Fall 2014)**.** Show that the equation $5^x + 6^x = 13^x$ has exactly one real solution.

Hint. Divide the left-hand side by the right-hand side.

Problem 6.3 (Spring 2014)**.** Let $f(x) = x^3 + px + q$ for $x \in \mathbb{R}$ with $p, q \in \mathbb{R}$ and $p > 0$. Show that f has a unique real root.

Problem 6.4 (Fall 2013)**.** (a) Show $\displaystyle\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$.

(b) Find the value A of the following improper integral:

$$A = \int_1^\infty e^{-x^2+2x+5}\,dx.$$

Problem 6.5 (Spring 2012)**.** Evaluate

$$\lim_{n\to\infty}\left(\frac{1}{n}\sum_{k=1}^{n}\sin\left(\frac{k\pi}{2n}\right)\right).$$